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Investigation of wave interaction with partially submerged wave screens

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ABSTRACT

Wave screens are thin vertical barriers composed of slotted rectangular elements with uniform gaps and fixed submergence, used to attenuate wave energy in sheltered basins such as marinas and small harbors. In this study, a combined laboratory and OpenFOAM based numerical investigation is conducted to quantify wave reflection, transmission, energy dissipation, and wave induced forces for partially submerged wave screens with varying porosity and submergence. An analytical formulation is further developed to estimate forces and clarify governing mechanisms. Results show that: (i) wave reflection reaches nearly its maximum value when the screen is submerged by only about 20% of the water depth, whereas further submergence primarily increases forces with limited gains in reflection. (ii) Porosity variations strongly affect transmission and energy dissipation but have a comparatively minor impact on forces. (iii) Wave steepness governs both attenuation efficiency and force levels, with steeper waves producing higher dissipation and loads even at shallow submergence.

1. Introduction

Coastal infrastructure, small harbors, and sheltered areas such as lakes, bays, or lagoons are subject to wave action that can disrupt their functionality and tranquility (Lee, 1966; Mani and Jayakumar, 1995). In fact, during strong events or storms, docks and other protective structures may sustain damage or even fail, significantly reducing the durability of the infrastructure (Efstathopoulos and Balomenos, 2023). Protecting harbors from wave-induced damage is a pressing challenge, especially in regions with limited resources or affected by high vessel traffic (Scarpa et al., 2019). Damages can lead to costly repairs and affect commercial and recreational maritime activities (Abdelhafez et al., 2021). In unprotected areas, waves can enter marinas and harbors, hitting vertical walls in docks and quays, and amplifying the wave height up to 3.9 times (Knowles and Yeh, 2019). To address these challenges, it is essential to implement reliable, cost-effective solutions that can attenuate the incoming wave and limit wave reflection.

Typical solutions involve sloping rubble margins, floating absorbing docks, and absorbing quays (S. Hettiarachchi and Wallsop, 1989; Hopkins et al., 2022). Sloping rubble margins rely on their porous and inclined surfaces to dissipate wave energy through turbulence and internal friction (Vicinanza et al., 2019). These structures are effective across a broad spectrum of wave conditions and provide ecological benefits by creating habitats for marine organisms (Manny et al., 1985). However, sloping rubble margins require more physical space than the other two options, making them less practical inside sheltered areas. Furthermore, the costs associated with sourcing, transporting, and placing the rocks needed to build such structures can be prohibitively high (Neelamani and Al-Anjari, 2021).

Floating attenuating docks offer a versatile solution since they serve dual purpose by acting as wave attenuators (Cutajar et al., 2023). Their modular design allows for ease of installation, removal, and adjustment, making them ideal for temporary or evolving harbor configurations (Zhao et al., 2019). However, while floating docks are typically the best solution for new constructions or damaged docks, absorbing quays are a more cost-effective solution for undamaged existing docks (Ruggeri et al., 2019).

Among the various types of absorbing quays, wave screens are a widely used option (Li et al., 2022; Saad et al., 2023). wave screens are composed of wooden slats featuring a series of gaps, which dissipate wave energy through the formation of viscous eddies as water flows through them (Hee Min Teh et al., 2013). Unlike solid walls, wave screens are designed to significantly reduce reflected wave energy, enhancing navigational safety (Isaacson et al., 1998). Additionally,

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these structures offer several advantages with respect to solid walls, including improved water circulation within harbors, unobstructed fish passage, facilitation of sediment transport across the breakwater, and low construction costs (Mani and Jayakumar, 1995; Neelamani and Al-Anjari, 2021).

Wave screens share conceptual similarities with closely spaced pile breakwaters, which have been extensively studied in both Japan and China since the pioneering work of Hayashi et al. (1966; 1968). Closely spaced pile breakwaters attenuate wave energy through turbulence generated by wave-pile interaction, with pile spacing being the primary parameter governing transmission (Sathyanarayana et al., 2024; Peng et al., 2023). This concept was further developed into the curtain-wall-pile breakwater (CPB), which combines a solid upper wall with closely spaced piles in the lower portion (Suh et al., 2006, 2007). The CPB shares functional similarities with the partially submerged wave screen studied herein, although the wave screen relies on slotted gaps rather than a solid curtain wall to govern the transmission and reflection behavior. The key distinction is that pile breakwaters employ arrays of cylindrical elements, whereas the wave screens examined herein use a vertical platform with horizontal slats whose gap geometry governs the dissipation mechanism. One notable application of wave screen is in the Lagoon of Venice, Italy, where they protect a small craft marina from locally generated waves within the lagoon (Fig. 1). Here, the wave screen was deployed to both reduce wave reflection and maintain tranquility inside the marina. Additionally, partially submerged wave screen installations can be found in several locations worldwide, including Baie-Comeau Harbor and Chandler Harbor in Canada, Roscoff Harbor in France, Half Moon Bay Marina in New Zealand, and Plymouth Harbor in the United States.

Studies on wave screens have progressed from simple porosity-based estimates to detailed hydrodynamic and structural formulations. Wiegel (1960) first expressed transmission as $K_t = \sqrt{f}$ (where f is the porosity of the wave screen) although this formulation underestimates transmission by neglecting reflection. Subsequent work incorporated geometric effects more explicitly. Bennett et al. (1992) introduced a semi-empirical nonlinear dissipation term linked to porosity, removing the need for experimental calibration, while Kriebel (1993) derived expressions for transmission and forces for with varying porosity for a fully submerged wave screen. Isaacson et al. (1998) extended the theory using a porous parameter and eigenfunction expansion to compute transmission and reflection. Then, Bergmann and Oumeraci (2000) modified Goda's method (Goda Y., 1974) by incorporating porosity to predict pressures on porous walls like wave screens. Refinements continued with Tabet-Aoul and Lambert (2003) and Wiebe et al. (2014), who adapted Goda's formula for porous walls with a backing berm or for elevated nonporous structures, respectively, and Koraim et al. (2011), who verified eigenfunction-based predictions for double horizontal slotted barriers through laboratory tests. Koraim (2014) further investigated pile-supported L-shaped bars functioning as a screen breakwater, demonstrating the effectiveness of hybrid pile-screen configurations in reducing transmission. Finally, Li et al. (2022) examined how relative draft and spacing influence the performance of twin wave screens through laboratory experiments. They characterized transmission, reflection, and run-up and proposed an empirical relation for transmission, with a focus on resonance that can develop in the region between the two screens. More broadly, studies on porous and flexible membrane breakwaters have demonstrated that submergence depth and membrane porosity are key parameters governing transmission, reflection, and dissipation coefficients, findings that are consistent with and complementary to those reported here for rigid slotted wave screens (Guo et al., 2022, 2023, 2020).

Other studies examined how wave screens design affects pressure drop and forces. Neelamani and Al-Anjari (2021) demonstrated progressive pressure reduction across multiple slotted barriers in random seas. Chanda and Pramanik (2023) showed that barrier spacing, for multiple wave screens, strongly influences load reduction. More recent work compares specific geometries; for example, Kumaran et al. (2025) reported that the orientation of the gaps of wave screens, whether horizontal or vertical, does not affect the pressure distribution, and that the highest pressures occur near the mean water level. Kakuno et al. (1993) analyzed small-amplitude waves interaction with a wave screen made of cylindrical elements with a back wall, while Sahoo et al. (2000) found that oblique incident waves exert lower dynamic pressures compared to normal incident waves. Under random waves, Krishnakumar (2010) observed a pressure drop of 12% for a single screen and 17% for a double screen relative to a bare wall, in general with lower total pressure in deep water compared to shallow water conditions. Poguluri and Cho (2020) further applied a matched eigenfunction expansion to regular waves interacting with wave screen of different porosities and showed that partial reflection can locally amplify free surface oscillations near the screen, with implications for confined basins.

The submergence depth of the wave screen is critical to its design, since it is expected to change the force acting on the screen. However, despite the existing body of literature, there is still insufficient data to determine how forces, wave reflection, transmission and energy dissipation vary with submergence depth. In fact, by decreasing submergence depth, the forces acting on the structure and wave reflection can be decreased, resulting in a more durable and economic design and better protection of the offshore navigation sites. The reduction in submergence of the wave screen reduces the material needed for the construction and reduces the loads, enhancing structural stability.

We address this gap by conducting a combination of physical experiments and numerical simulations across various wave fields and geometrical wave screen configurations to evaluate their impact on

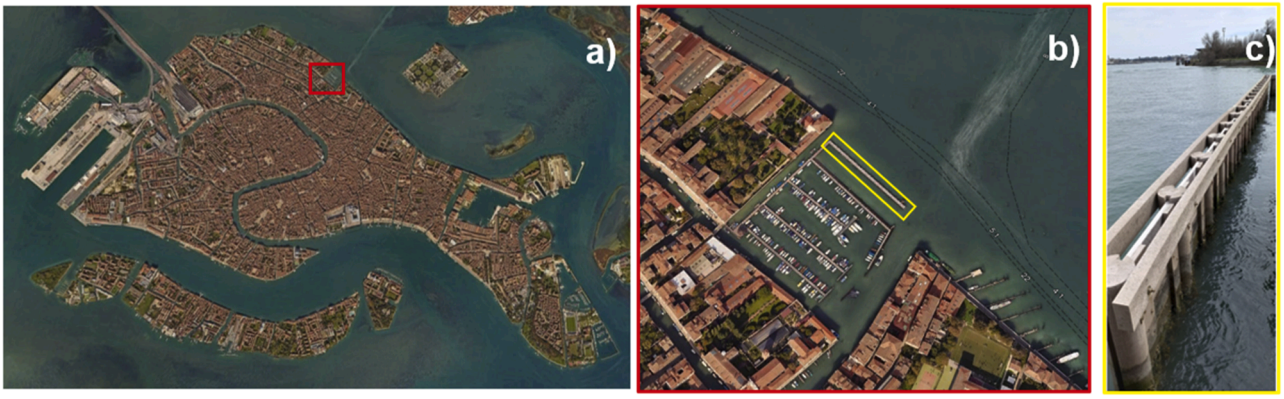


Fig. 1. a) Venice, Italy. The red square marks the location of Marina "Sacca della Misericordia". b) Marina area with heavy vessel traffic. Public transport routes are shown by a black dashed line. The partially submerged wave screen absorbing quay is marked in yellow. c) Detail of the wave screen absorbing quay.

wave forces, wave transmission, wave reflection and energy dissipation. First, a set of physical experiments were conducted for specific geometry under several wave conditions. However, because it is impractical and time-consuming to test multiple wave conditions across various geometries in a flume, we calibrated a numerical model using the data from these experiments. We then used this model to explore additional scenarios beyond the limited parameter space of the physical testing. Finally, we provide an analytical formulation to calculate wave transmission, reflection, energy dissipation and forces on a wave screen with varying submergence, gap width, and wave field.

This paper is organized as follows. Section 2 describes the methods, including the experimental set-up (Section 2.1) and the OpenFOAM numerical model, covering model details and the numerical wave flume configuration (Section 2.2). Section 3 presents the results, beginning with a validation of the numerical model against experimental data for a fully submerged wave screen (Section 3.1), followed by a parametric numerical investigation of the effects of submergence, porosity, and wave steepness on wave reflection, transmission, energy dissipation, and forces (Section 3.2), and a comparison between numerical results and the analytical theory (Section 3.3). Section 4 develops the analytical formulation based on potential flow theory. Section 5 discusses the implications of the findings, and Section 6 summarizes the main conclusions.

2. Methods

2.1. Experimental set-up

Physical model tests are conducted in a wave flume at the University of Padua (Italy), which is 36.0 m long, 1.0 m wide, and 1.30 m deep. The flume is equipped with a piston wavemaker capable of generating both regular and irregular waves, with active wave absorption. Thin metal plates, arranged parallel to the side walls, are placed in front of the wavemaker to prevent transverse oscillations. The bed is made of lean concrete, the sides are made of glass, and the flume is equipped with a recirculation system. In this study, the piston mode is used, and the model bottom was fixed, horizontal, and non-erodible. A slotted wave screen (Fig. 2), consisting of nine wooden planks (9 cm wide and 1 m high), is positioned 7.5 m from the wave generator (Fig. 3). Each plank is separated from the adjacent plank by $s = 2.1$ cm wide slits, for a total of 8 slits. The end planks are separated from the glass boundary by 1.05 cm wide slits, which results in an overall porosity P of 0.19. The barrier is supported by a metallic structure at the rear, with submergence reaching its maximum value ($a/h = 1$). Two arrays of four wave gauges were used to measure incident and reflected waves in both offshore and inshore sides (Fig. 4), with a sampling frequency of 100 Hz. The first gauge of the offshore array is positioned 5.77 m from the wave generator, while the first gauge of the nearshore array is placed 11.3 m away. In both arrays, the spacing between wave gauges are, from offshore to inshore, 19 cm, 11 cm, and 53 cm, as shown in Fig. 4. Twelve pressure gauges (sampling frequency 1200 Hz) are installed along the central wooden plank, with the first located 12.5 cm from the bottom and the remaining gauges spaced 5 cm apart, as shown in Fig. 3c. To counter wave reflection caused by the vertical wall, consolidated active wave absorption from HR Wallingford is used.

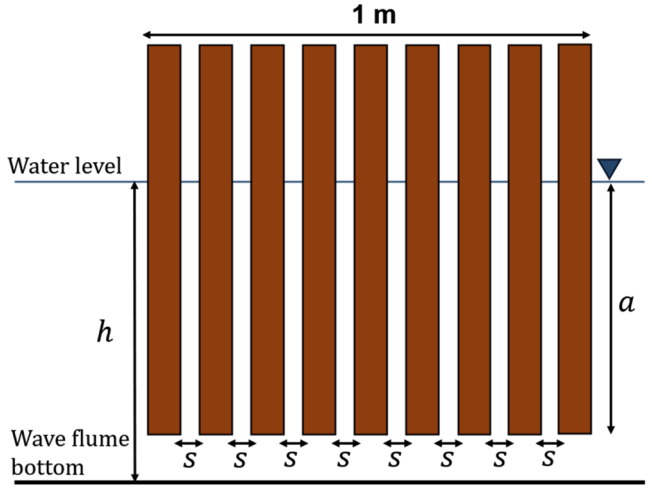


Fig. 2. Schematic representation of the partially submerged wave screen (front view). The width of the screen is 1 m $\boldsymbol{a}$ represents the screen submergence (takes values from $-\boldsymbol{h}$ to 0), $\boldsymbol{s}$ is the distance between two wood panels (i.e. gap width), $\boldsymbol{h}$ is the water depth. The blue horizontal line represents the still water level.

The comparison between numerical and experimental investigations is conducted on 39 regular wave cases. The scenarios include waves with heights ranging from 0.05 m to 0.15 m, periods between 0.8 s and 2 s, and three different water depths (0.5 m, 0.6 m, and 0.7 m). The surface elevations recorded during the experiments are band-pass filtered in the range of 0.1–20 Hz to remove the mean and possible noise. Incident and reflected wave components were separated using a multi gauge least squares method based on linear wave theory. The free surface elevation at each gauge was represented as the superposition of incident and reflected waves with known phase shifts determined by the gauge spacing. The complex amplitudes of the incident and reflected components were obtained by solving an overdetermined system in a least squares sense using the four wave gauges, following Zelt et al. (1993). While the method is grounded in linear wave theory, Zelt and Skjelbreia (1992) demonstrated that reliable separation can be achieved under moderate nonlinearity if gauge spacing is selected appropriately and that wave energy is concentrated at the fundamental frequency. The wave conditions examined in this study satisfy these criteria, as nonlinearity levels characterized by wave steepness and relative wave height remain within the applicable range of the method. Finally, the transmission coefficient K_t is evaluated as the ratio of the transmitted wave height to the incident wave height.

2.2. Numerical model

A numerical wave flume is used with the OpenFOAM (Weller et al., 1998) toolbox waves2Foam, which numerically solves the three-dimensional Reynolds-Averaged Navier-Stokes (RANS) equations for an incompressible fluid. The model employs the finite volume method (FVM), dividing the domain into a finite number of cells and calculating flow variables at their centroids. This approach was chosen to accurately reproduce non-linear effects, including vortex formation and complex fluid-structure interactions.

2.2.1. Model details

The momentum and continuity equations solved by the OpenFOAM model (Weller et al., 1998) read:

$$\frac{\partial \rho \boldsymbol{u}}{\partial t} + \nabla \cdot \rho \boldsymbol{u}\boldsymbol{u}^T = \; -\nabla p + \boldsymbol{g} \cdot (\boldsymbol{x} - \boldsymbol{x}_r)\nabla \rho + \mu_{tot} \nabla^2 \, \boldsymbol{u}, \tag{1}$$

$$\nabla \cdot \boldsymbol{u} = 0, \tag{2}$$

where ρ is the density of the fluid, t is time, $\boldsymbol{u}$ is the velocity vector in Cartesian coordinates, $\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right)$ is the differential operator, p is the excess pressure, $\boldsymbol{g}$ is the vector acceleration due to gravity, $\boldsymbol{x} = (x, y, z)$ is the Cartesian coordinate vector, $\boldsymbol{x}_r$ is a reference location (here defined as the sea level) and μ_{tot} is the total dynamic viscosity. The excess of pressure is defined as $p = p_{tot} - \rho g \boldsymbol{x}$, where p_{tot} is the total pressure.

The interface of water and air is captured using the VOF method (Hirt and Nichols, 1981). This method divides the water and air phase by

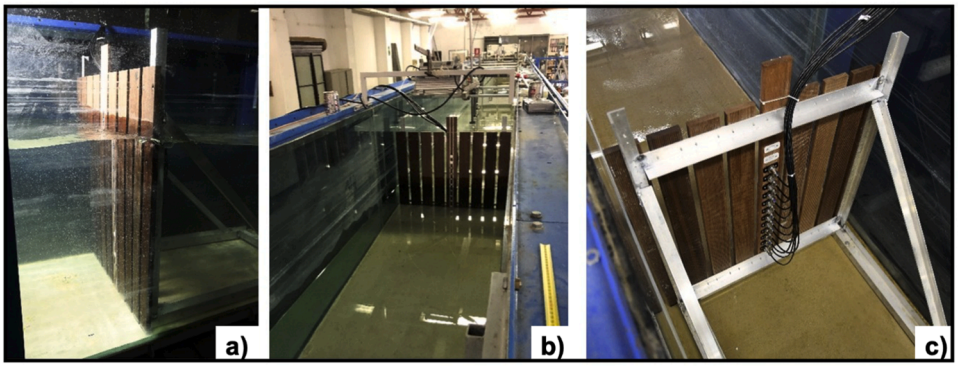


Fig. 3. Side view a), front view b) and rear-view c) of the wooden wave screen tested in the wave flume of Padova University. As detailed in c), 12 pressure gauges are placed on the central wooden panel.

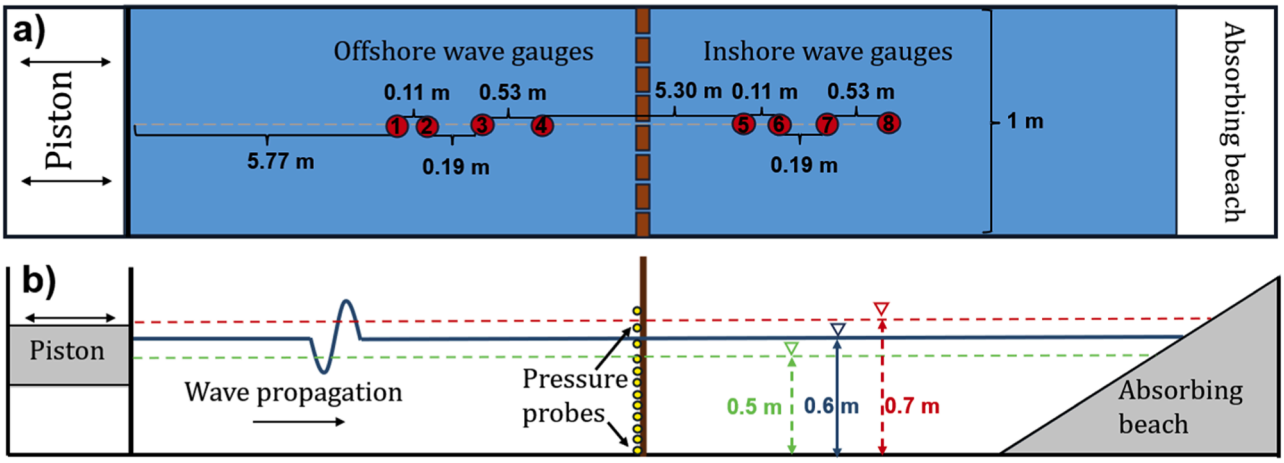


Fig. 4. (a) upper view and (b) side view of the experimental wave flume scheme. The location of the eight gauges, in meters, is referenced from the piston. . The wave screen is indicated in brown, in the center of the domain. In (b), the blue line, green and red dashed lines represent the water levels considered in the experiments. Waves propagate from the left (offshore) to the right (inshore) side.

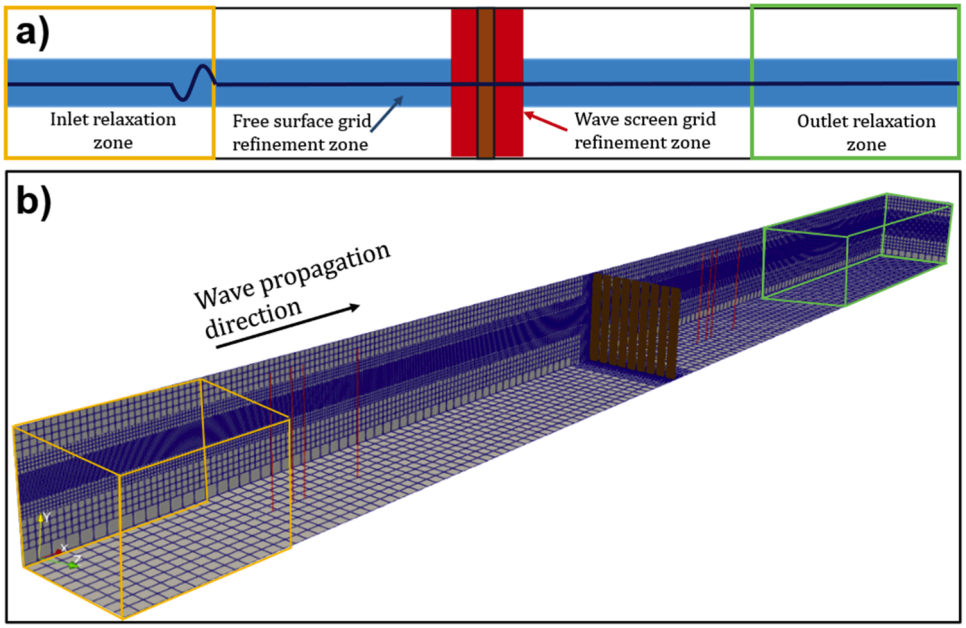


Fig. 5. a) Numerical grid with two refinement zones: near the free surface (light blue) and near the wave screen (red). The yellow region indicates the wave generation and green region absorption relaxation zones in both figures. b) Unstructured grid of the numerical wave flume. Red lines indicate probe locations for wave reflection and transmission estimates. Waves propagate from left to right; the wave screen is at the center of the domain.

defining a variable α, which provides a fractional volume of water for each of the two fluids in the cell. To avoid excessive damping of the wave at the water-air interface, a buoyancy modified $k-\omega$ SST turbulence model was used (Devolder et al., 2018). Here, a buoyancy term is added to the turbulent kinetic energy (k) equation to avoid the large production of k due to the unbalanced pressure gradient and density gradient between water and air at the free surface. Pressure–velocity coupling is handled with the PIMPLE algorithm, using GAMG for the pressure equation and a second-order implicit Crank–Nicolson scheme for time discretization (Weller et al., 1998).

2.2.2. The numerical wave flume

The numerical wave flume consists of an inlet relaxation zone spanning 1.5 times the wavelength L, and an outlet relaxation zone extending 2 times the wavelength, and it replicates the area between the first and last wave gauge in the lab (see Section 2.1). The computational mesh is refined to a high-resolution zone near the free water surface, set to 20 cells per wave height H with square cells (Aspect Ratio = 1). A convergence study was performed for one case progressively increasing from 2 to 20 cells per wave height to reach convergence in the free surface profiles. The results confirmed that 20 cells per wave height were needed, thus confirming what recommended by Windt et al. (2019). A finer region around the wave screen is further refined 4 times using snappyHexMesh (red area in Fig. 5a). Finally, the background mesh provides the base resolution for the domain and is obtained by derefining the water surface mesh by a factor of 8. In the numerical domain, wave absorption and generation are performed with the wave relaxation algorithm presented by Jacobsen et al. (2012). The modeled wave screen and bottom are set as non-slip conditions. Constant pressure is enforced at the atmosphere boundary which allows the air to flow in and out while restricting the water to outflow only. Spalding wall functions are employed to approximate the turbulence in the near-wall region, with the first cell center placed in the log-law region to ensure accuracy. The y^+ values of the simulations are in the range of $y^+ \in [30, 300]$. This range is optimal as it allows the use of wall functions while avoiding the buffer layer, ensuring an accurate representation of near-wall turbulence without the need for extremely fine mesh resolution near the wall. Each simulation is run for 20 wave cycles (20 T) and the CPU time increases for wave fields with higher period T. This is because the dimensions of the numerical wave flume are adapted to the wavelength (proportional to T^2). OpenFOAM runs in parallel via the Message Passing Interface (MPI) on the HiperGator high-performance Linux cluster using 32 cores (3.3 GHz) to increase computational speed.

The primary advantage of parallelization is the ability to run multiple simulations with different wave screen configurations simultaneously. By creating 3D model files of the wave screen and integrating them into a validated numerical wave flume, a wide range of wave fields, structure submergence depths, and gap width are simulated. Experimentally replicating these scenarios would be prohibitively expensive and time-intensive, as it would require constructing physical models, extensive wave tank testing, and labor-intensive data collection.

The wave screens are treated as rigid barriers constructed from structural materials typical of coastal installations, such as steel or engineered composites, with sufficient stiffness that elastic deformation under the tested wave conditions remains negligible relative to the structural dimensions (Aristodemo et al., 2021).

Firstly, the model is used to reproduce the experimental flume design with the same 39 wave conditions. Next, the model is run for depth $h = 0.6$ m, using 6 different wave fields among the validated ones (including with $H = 0.05$ m, 0.10 m, 0.15 m and $T = 1$ s, 1.5 s, 2 s). The dimensionless draft of the structure (a/h) varies from 0 to 1. Hereinafter, the porosity P of the wave screen is defined as the ratio of void area (which is number of gaps times the gaps width s, with the number of gaps fixed to eight) with respect to the total area, and three porosity values (0.05, 0.10, 0.20) are tested. The thickness of the wave screen is fixed at 0.02 m, matching the experimental setup. This comprehensive approach results in a total of 126 combinations explored (6 wave fields × 7 draft ratios × 3 porosity values), providing a rich dataset for analysis.

2.3. Hydrodynamic theory

This section aims to quantify the wave transmission, reflection, energy dissipation and forces using linear wave theory as an expansion of the work of Mei (1989) and Kriebel (1993) to conditions where the wave screen has variable submergence. The analytical approach follows a first-order hydraulic idealization and is intended to capture the dominant mechanisms governing wave–screen interaction under regular wave conditions, rather than to provide a full potential-flow solution.

Energy dissipation during flow expansion is assumed to follow the same loss behavior as steady-flow contraction (Garde, 1997). The calculation uses only the horizontal velocity component which reaches its maximum during the crest and trough phases of the wave because those phases coincide with the largest pressure gradients across the wave screen (Lyons, 2018). These conditions therefore define the moment of highest wave force and provide a meaningful basis for estimating wave dissipation which depends on both reflection and transmission (van Gent et. al, 2023).

We neglect energy dissipation during flow contraction. Additionally, it is assumed that the mass and momentum conservation equations can be applied first over a horizontal layer of thickness dz, with z the vertical coordinate, and then integrated across the depth.

We consider the flow along the centerline of two adjacent panels of the wave screen (see Fig. 6). Conservation of mass requires that the mass flux at each of the three sections "$i+r$", "i", and "r" is equal. The formulation is schematic in nature and assumes that the characteristic distances between sections (i + r), c, and t are small relative to the wavelength, such that phase differences between the incident, reflected, and transmitted velocity components can be neglected to first order. Therefore, by assuming linear superposition of the incident and reflected waves, flow velocities can be written as:

$$U_i + U_r = \mu P U_c = U_t, \tag{3}$$

where the subscript i, r and t represent, respectively, the incident, reflected and transmitted wave, $\mu = 0.63 + 0.37 P^3$ is the contraction coefficient given by the Weisbach formulation (Oertel, 2004) and P is the porosity of the wave screen. Based on linear wave theory, the velocity distribution along the vertical axis z for the incident, reflected, and transmitted of given H and T with depth h can be written as follows:

$$\left|U_{i,r,t}(z)\right| = \omega \frac{H_{i,r,t}}{2} \frac{\cosh[k(h+z)]}{\sinh(kh)}, \tag{4}$$

where z is measured from the still water level, k is the wave number, h is the water depth, ω is the angular wave frequency, and the subscript i, r, t defines whether U and H are representative of the incident, reflected or transmitted wave, respectively.

By replacing Eqs. (4) into (3), we have that:

$$H_i - H_r = H_t . \tag{5}$$

Thus, the reflection ($K_r = H_r/H_i$) and transmission coefficient ($K_t = H_t/H_i$) are related as follows:

$$K_t = 1 - K_r, \tag{6}$$

Note that this relationship is a direct consequence of mass conservation only when assuming that the unsteady flow can be represented as a sequence of steady states. Additionally, we define the wave energy dissipation coefficient (K_e) as:

$$K_e = 1 - K_t^2 - K_r^2 \tag{7}$$

The pressure gradient across the wall is determined by applying the

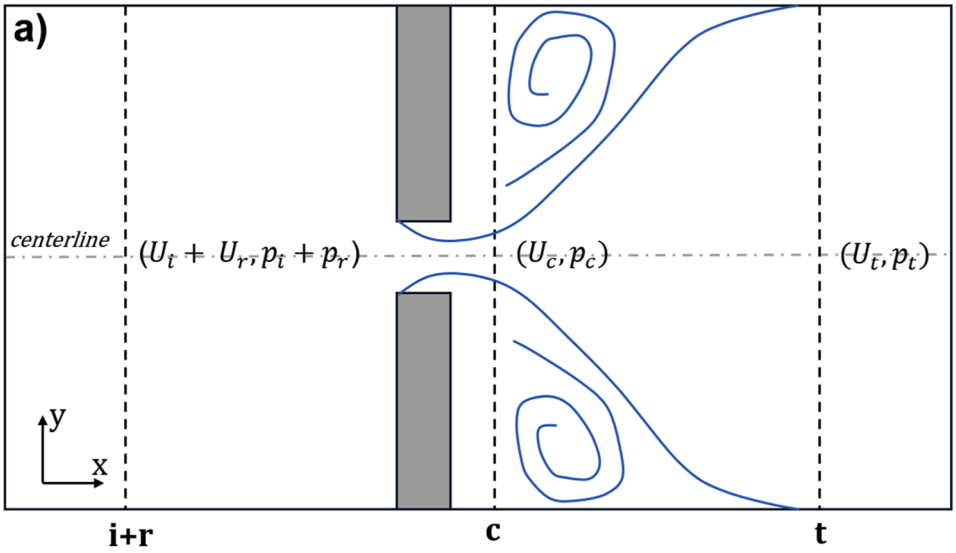


Fig. 6. Scheme of the flow through the gaps in a wave screen. Section "$(i + r)$" is placed in front of the screen, where are present only incident and reflected waves. Section "c" is placed right after the gap, where the flow is contracted. Section "t" is set far from the screen, where the transmitted wave propagates.

momentum and energy equations. The conservation of momentum is applied between sections "c" and "t" in Fig. 6 and reads:

$$p_c - p_t = \rho U_t (U_t - U_c), \tag{8}$$

Where p_c is the excess pressure in the section of contracted flow, and p_t is the excess pressure of the transmitted wave (all pressures mentioned henceforth are excess pressures, and we use the term "pressure" for brevity). By using the right equality in (3), reads:

$$p_c = p_t - \rho U_t^2 \left(\frac{1}{\mu P} - 1 \right). \tag{9}$$

The energy conservation between sections "$i + r$" and "c", assuming linear superposition of the incident and reflected waves, and negligible energy dissipation during the flow contraction, reads:

$$(p_i + p_r) + \frac{\rho}{2}(U_i + U_r)^2 = p_c + U_c^2, \tag{10}$$

which allows us to relate the incident and reflected pressures to the pressure in the contraction. Combining Eqs. (9), (10), (3) and defining a contraction function $f = (1/\mu P - 1)^2$ finally allows to compute the pressure drop as:

$$(p_i + p_r) - p_t = \frac{1}{2} f \rho U_t^2. \tag{11}$$

The dynamic pressure distribution along the vertical axis z at any layer of fluid given by linear theory of a wave with given H and T, and depth h is:

$$|p_{i,r,t}(z)| = \rho g \frac{H_{i,r,t}}{2} \frac{\cosh[k(h+z)]}{\cosh(kh)}, \tag{12}$$

where h is the water depth. By substituting the wave pressure and velocity distributions in Eq. (11), using Eq. (5) and integrating over the structure submergence $= [-a, 0]$, we derive the following equation:

$$(1 - K_t) \left\{ \frac{\sinh(kh) - \sinh[k(h+a)]}{\cosh(kh)} \right\} = \frac{f \omega^2 K_t^2 H_i}{2g} \left\{ \frac{-2ka + \sinh(2kh) - \sinh[2k(h+a)]}{4k \sinh^2(kh)} \right\}. \tag{13}$$

To simplify the expression, we define the function σ_t as:

$$\sigma_t = \frac{f}{8} \frac{H_i \omega^2}{gk} \left\{ \frac{-2ka + \sinh(2kh) - \sinh[2k(h+a)]}{\sinh^2(kh) \frac{\sinh(kh) - \sinh[k(h+a)]}{\cosh(kh)}} \right\}, \tag{14}$$

which is a function of the incident wave height H_i, and includes the depth h, submergence a, porosity P (considered in f), and the wave period T, as it includes the wave number k and ω (angular frequency). Consequently, Eq. (13) reads:

$$\sigma_t K_t^2 + K_t - 1 = 0, \tag{15}$$

which can be solved for the unknown K_t:

$$K_t = \frac{-1 + \sqrt{1 + 4\sigma_t}}{2\sigma_t}. \tag{16}$$

To compute K_r and K_e we can substitute Eq. (16) in Eqs. (6) and (7). Then, the wave force F per unit width is calculated as the difference of the pressure between distribution of incident $p_i(z)$ and reflected wave $p_r(z)$ (offshore side) and transmitted wave $p_t(z)$ (inshore side) in each horizontal layer of thickness dz. Integrating from the free surface to the submergence a, and multiplying by the effective area of the screen $(1 - P)$, we obtain:

$$F = \int_a^0 (p_i(z) + p_r(z) - p_t(z))(1 - P) dz$$

$$= \int_a^0 \rho g H_i \frac{\cosh[k(h+z)]}{\cosh(kh)} (1 - K_t)(1 - P) dz. \tag{17}$$

The integral reads:

$$F = \rho g \frac{H_i}{k} \frac{\sinh(kh) - \sinh[k(h+a)]}{\cosh(kh)} (1 - K_t)(1 - P), \tag{18}$$

which computes the force exerted by a wave on a wave screen of given porosity P and submerged a below the water surface. If the wall is solid and fully submerged (i.e. $P = 0$, $K_t = 0$, $a = -h$), Eq. (18) allows to compute the force of a linear wave of given wave height H and wave number k on a solid wall F_s as in Kriebel (1993):

$$F_s = \rho g \frac{H_i}{k} \tanh(kh) \ . \tag{19}$$

3. Results

3.1. Comparison between numerical and experimental model (full submerged wave screen)

The numerical results are compared to the experimental ones to check the ability of the numerical model to reproduce the wave screen behavior. To assess the quality of the numerical results in terms of incident, reflected and transmitted wave heights, we use two performance metrics: the coefficient of efficiency *NSE* (Nash and Sutcliffe, 1970) and the index of agreement *D* (Willmott et al., 1985):

$$NSE = 1 - \frac{\sum (y_N - y_E)^2}{\sum (y_E - \overline{y_E})^2} \tag{20}$$

$$D = 1 - \frac{\sum (y_N - y_E)^2}{\sum |(y_N - \overline{y_E})^2| + |(y_E - \overline{y_E})^2|} \tag{21}$$

Complete disagreement is described by $D = 0$ and negative *NSE*. All indexes are equal to 1 for perfect agreement. The two indices were evaluated based on the values of incident, transmitted, and reflected wave height, as calculated for the 39 experimental and numerical tests. For incident waves, NSE = 0.978 and D = 0.995; for transmitted waves, NSE = 0.884 and D = 0.976; and for reflected waves, NSE = 0.995 and D = 0.999. The comparison between the surface elevations measured at the eight wave gauges (located as illustrated in Fig. 5) during the laboratory tests and the numerical simulations (with parameters set at H = 0.1 m, T = 1.5 s, and h = 0.5 m) is shown in Fig. 7. Peak amplitude differences average of approximately 4% for offshore gauges and 6.5% for inshore gauges, confirming that the model correctly predicts wave energy distribution across the domain. Root mean square errors are 0.0086 m and 0.0061 m for offshore and inshore gauges respectively,

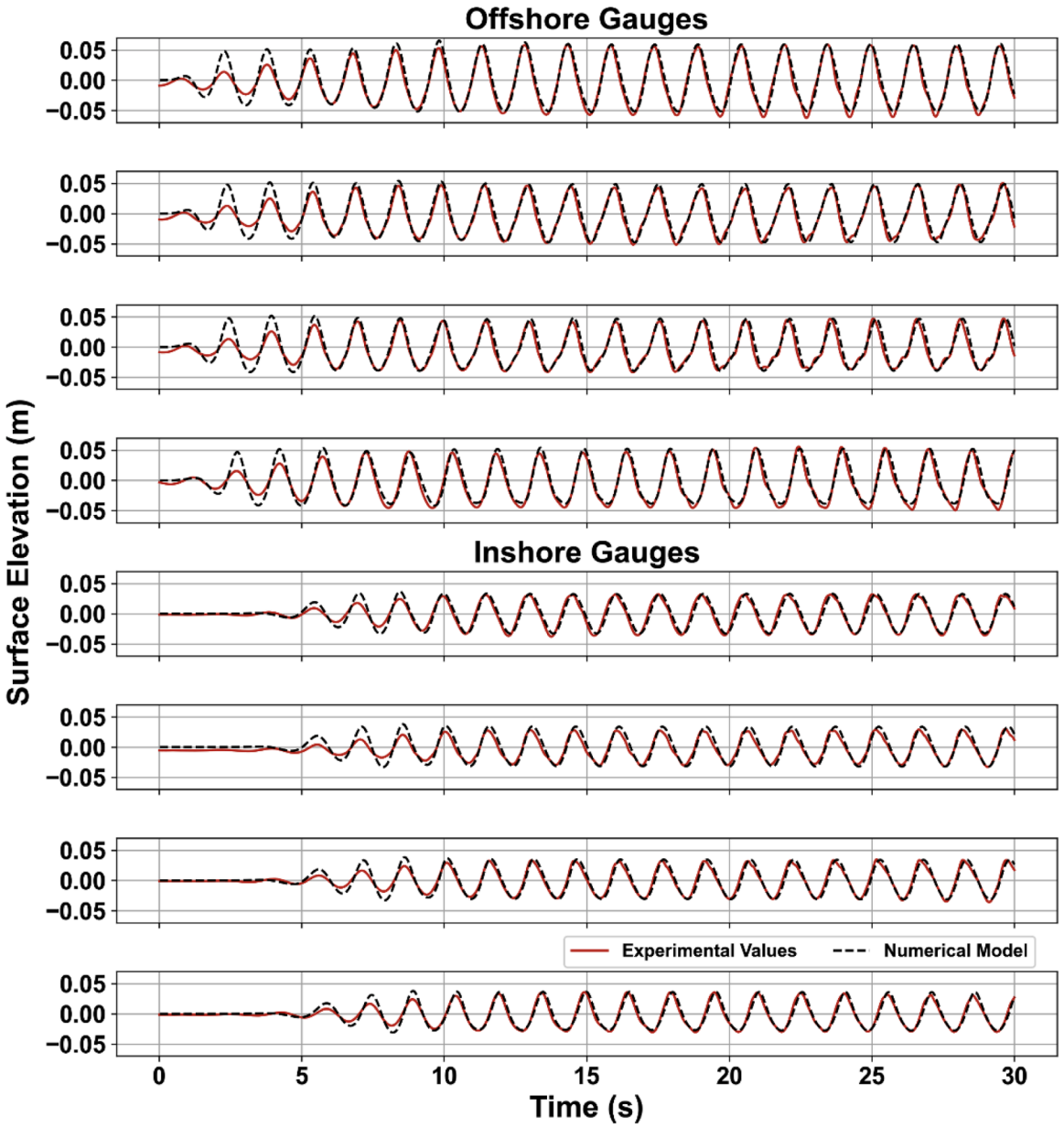


Fig. 7. Comparison between the 8 wave gauges in the wave flume and the numerical results. The subplots are order from offshore (first gauge, at the top) to inshore (last gauge, at the bottom), whose location is shown in Fig. 5. The wave field considered in this plot is wave height (H) = 0.1 m, period (T) = 1.5 s, and depth (h) = 0.5 m.

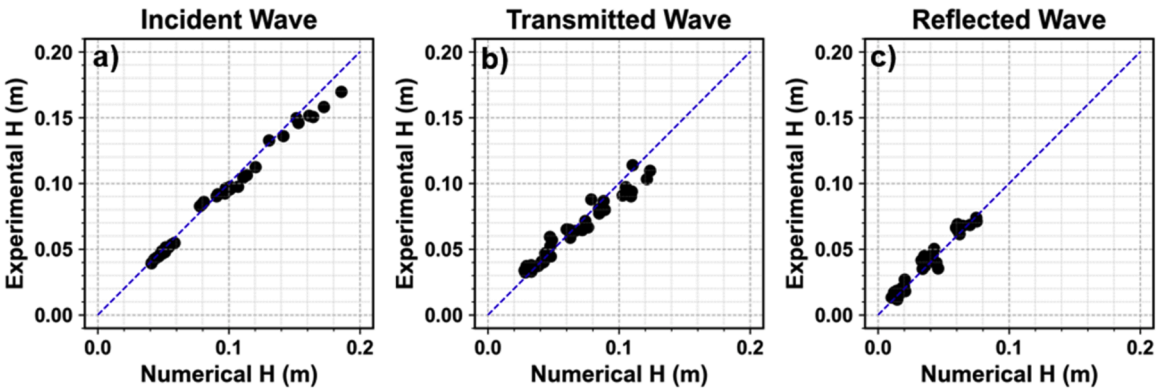


Fig. 8. Comparison between numerical and experimental wave heights. (a) Incident wave. (b) Transmitted wave downstream of the wave screen. (c) Reflected wave upstream of the wave screen.

demonstrating that absolute deviations from measured values remain small relative to the incident wave heights tested.

Fig. 8a, b and c present the experimental incident, transmitted and reflected wave heights H versus the numerical ones for the 39 tests, respectively. Experimental H values of the incident waves are plotted against their numerical counterparts, enabling evaluation of whether the waves emerging from the relaxation zone match the prescribed conditions. Finally, the transmitted and reflected H from experiments are plotted versus numerical results, to verify that the energy dissipation process is correctly captured.

Fig. 9 shows the dimensionless pressure distribution $p/\rho gz$, where z is the elevation with respect to the bottom of the flume, along the slotted wave screen for different wave conditions, comparing numerical and experimental results at the time of maximum integrated pressure. This comparison verifies that the model computes force correctly. The numerical results reproduce the pressure distribution very close to the measured data, including the peak dynamic pressures near the still-water level.

Overall, the model reproduces incident, transmitted, reflected wave heights and pressure distributions with good accuracy, indicating it is suitable for analyzing wave screens interaction with different wave fields. In the next section, we will use the model to extend the parameter

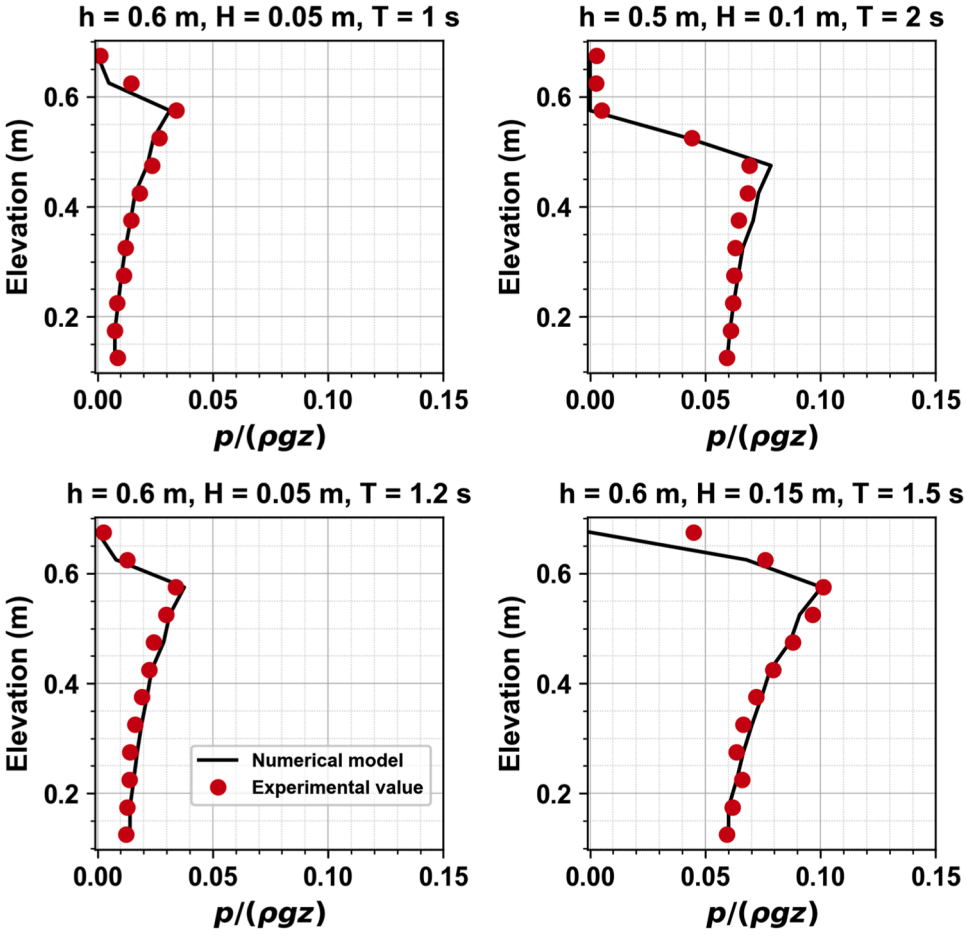


Fig. 9. Numerical versus experimental dimensionless pressure. The black line shows the vertical pressure profile at the instant of maximum integrated pressure from the model, and the red dots represent the corresponding maximum pressure values measured in the experiments.

space of relative draft and porosity.

3.2. Numerical model results

3.2.1. Wave transmission, reflection and energy dissipation

Wave transmission K_t, reflection K_r and energy dissipation K_e are the three coefficients used to evaluate the interaction between the wave screen and the wave field. Fig. 10 shows respectively K_r, K_t, and K_e as a function of the relative draft a/h for varying porosity and wave conditions. All the coefficients vary significantly with relative draft, porosity, and wave characteristics. As waves interact with the screen, a portion of their energy is reflected offshore, while some is transmitted through the slots or dissipated following the balance in Eq. (7).

In general, Wave screens with lower porosity and higher relative draft a/h generally exhibit higher energy dissipation K_e and lower transmission coefficients (K_t). However, wave reflection exhibits a plateau behavior. In fact, all the cases reach 80–90% of the maximum reflection while engaging only 0.2 relative draft of the structure. Because reflection indicates how much wave energy returns offshore, it remains a key metric to take in account. When reflection stays nearly constant while transmission decreases with varying draft of the wave screen, as seen in Fig. 10, the resulting increase in energy dissipation benefits the inshore side of the screen by reducing transmitted wave energy without causing a rise in reflected energy.

The wave characteristics also influence the wave energy balance. This is evident when comparing scenarios with the same wave height ($H = 0.15$ m) but varying periods ($T = 1.5$ s or 2 s, respectively orange and light green lines in Fig. 10). The higher period wave displays less reflection and higher transmission. As a result, Eq. (7) implies greater wave energy dissipation for the longer period case. When waves have the same period, but higher wave height, they tend to dissipate more energy. For example, at $T = 2.0$ s, K_e increases as H rises from 0.05 m to 0.15 m, showing greater energy dissipation for higher-amplitude waves, with the effect stronger for larger submergence. This follows the mechanism in Section 2.3: dissipation scales with the square of velocity, and crest velocity scales with wave amplitude (Phillips, 1977), so higher amplitude waves produce higher dissipation, hence reducing transmission in the inshore side.

The effect of wave steepness H/L on wave energy dissipation can also be examined considering the highest wave steepness case, ($H = 0.1$ m, $T = 1.0$ s, brown line in Fig. 10) which consistently shows the lowest K_t values across all the a/h values and therefore the highest wave energy dissipation.

Slotted wave screens dissipate wave energy primarily due to the increased water velocity in the gaps, along with the consequent vortex formation and dissipation through viscous effects, as shown in Fig. 11. The porosity of the screen influences how waves interact with the structure, modifying the behavior of the vortexes generated in the inshore side of the screen and the velocity inside the gaps. It has been shown that, for a fixed porosity, the number of gaps significantly affects the overall dissipation (Biondi, 2023). In general, screens with higher porosity dissipate less energy, because they allow a higher portion of the wave energy to be transmitted while having comparable reflection to smaller porosity.

The flow patterns shown in Fig. 11 illustrate how wave conditions influence the jet-like flow through the gaps and the associated vortex

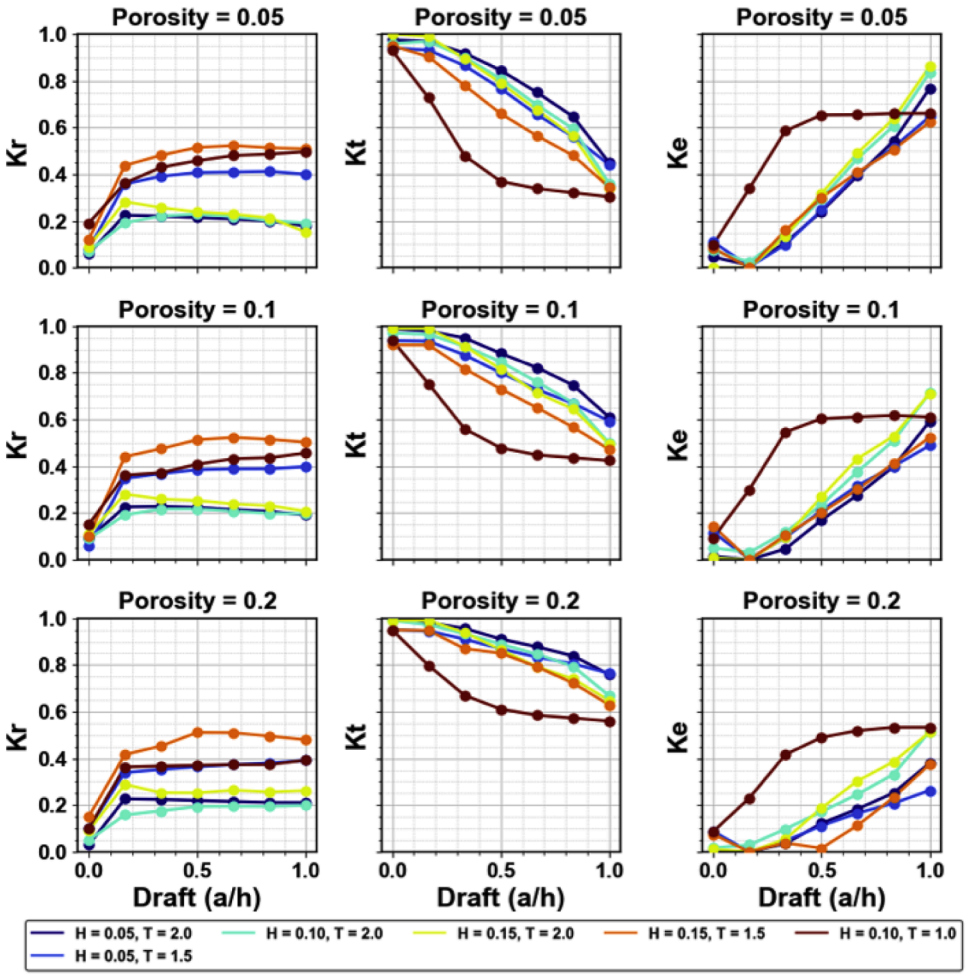


Fig. 10. Transmission, reflection and energy dissipation coefficients for different porosities, relative draft and wave fields.

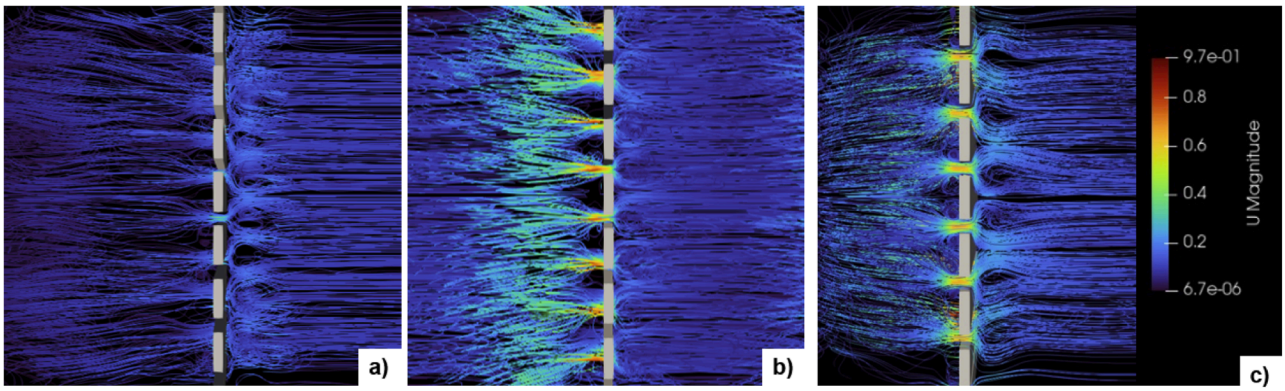


Fig. 11. Flow path through the gaps during the wave crest phase computed with the numerical model for a) $H = 0.05$ m, $T = 2$ s; b) $H = 0.1$ m, $T = 1$ s and c) $H = 0.1$ m, $T = 1.5$ s. Waves flow from the right side of the picture (offshore) to the left side (inshore). The color map represents the magnitude ($\sqrt{U_x^2 + U_y^2 + U_z^2}$).

structures on the inshore side. For longer wave periods (Fig. 11a), flow velocities through the gaps are lower and the vortices on the leeward side are weaker, resulting in less energy dissipation and therefore higher K_t and lower K_r. For shorter periods and higher steepness (Fig. 11b,c), the increased orbital velocities intensify the jet flow through the gaps, promoting stronger vortex formation and greater viscous dissipation. This mechanism drives the reduction in K_t observed at higher wave steepness in Fig. 10, while K_r remains comparatively stable because reflection is governed primarily by the solid projected area of the screen rather than by the local flow dynamics within the gaps. The limited sensitivity of K_r to porosity changes, also visible in Fig. 10, is consistent with this interpretation: reducing porosity increases the solid fraction and thus reflection, but the additional blockage has a comparatively minor effect on the vortex-dominated dissipation process that governs transmission.

3.2.2. Forces

Values for pressure over the wet surface of the wave screen (i.e. the portion of the structure that interacts with the wave) are extracted from OpenFOAM and integrated to find the force (F) per linear meter of structure. The force F considered is the mean of the ten highest forces recorded by the model over the simulation (i.e., the 10 instants with highest pressure gradient between inshore and offshore side of the wave screen – generally during the crest phase). These values are also compared to the forces obtained by the linear theory as described in Section 2.3. The force calculated by the numerical model is made dimensionless by dividing it by F_s, which is the force exerted by a linear wave with wave height H and period T on a fully submerged solid wall (e.g. porosity = 0), Eq. (19). This normalization isolates the influence of porosity and submergence on the force with varying wave field.

Fig. 12 shows the variation of dimensionless force (F/F_s) with relative draft (a/h) for different porosity values (porosity = 0.05, 0.1, and 0.2), and wave conditions. Across all porosities, F/F_s increases with the draft, mainly because of the increase in area of fluid-structure interaction. For the same reason, lower porosity values (e.g., porosity = 0.05) result in higher F/F_s for the same wave conditions, although these variations in porosity are not as pronounced. This is valuable for design, since lower porosity reduces wave transmission up to 35–40% and increases inshore protection while increasing the force by only about 5–10%. This is evident when comparing the transmission K_t in porosity of 0.2 and 0.05 in Fig. 10. Such increases in forces can be accommodated through structural design, while the enhanced attenuation of incoming waves provides additional protection to the sheltered area.

For a fixed wave height, decreasing the wave period tends to increase F/F_s closer to the surface and decrease it closer to the bottom. This occurs because shorter period waves behave as deep water waves, with orbital velocities concentrated near the surface and decaying rapidly with depth. Conversely, longer period waves behave more as shallow water waves, with orbital velocities distributed more uniformly over the water column, resulting in comparatively greater hydrodynamic loading on the lower portion of the wave screen. On the other hand, for a fixed wave period, the normalized force F/F_s exhibits a non-monotonic dependence on wave height. As H increases from 0.05 m to 0.1 m, F/F_s increases, but further increasing H to 0.15 m results in a decrease, while generally remaining above the $H = 0.05$ m case. Consequently, the largest $F/F_s$is obtained at the intermediate wave height $H = 0.1$ m rather than at the highest wave height tested. A notable case is the highest wave steepness H/L wave analyzed ($H = 0.1$ m; $T = 1$ s, brown line in Fig. 12), where the increase in F/F_s follows a different trend. In general, waves with higher wave steepness show a stronger decay of pressure with depth (see Fig. 9), which reduces loading near the base of

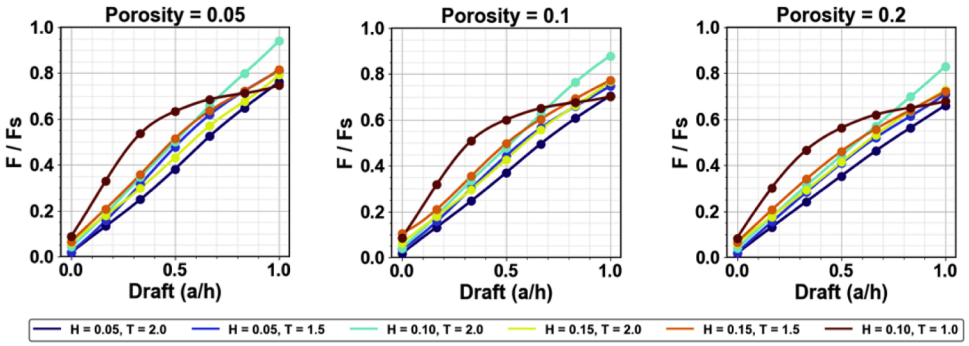


Fig. 12. Dimensionless forces for different porosities. The different colors represent different wave fields, identified by wave height H and period T.

the structure. For small drafts close to the surface, shorter period waves produce larger forces than longer period waves. As the draft increases, this trend reverses and shorter period waves generate lower forces than the longer period case. This affects not only the total horizontal force but also the overturning moment, since the center of pressure shifts upward toward the free surface. As a result, stresses on pile foundations increase due to the larger moment arm.

3.3. Comparison between numerical model and analytical theory

The analytical solution, based on linear wave theory and potential flow assumptions, provides an excellent estimation of the forces exerted by the waves on the structure among the wave field and geometrical configurations of the wave screen considered in this research. In general, the numerical model offers a more detailed depiction of wave–structure interactions by capturing nonlinear effects and local flow phenomena. Fig. 13 compares the force (F) calculated using the linear theory (see Eq. (18)) with the model outputs. Overall, the agreement between the numerical and analytical results is excellent.

Numerical and analytical differences stem from a simplification in the analytical derivation (Section 2.3), where energy dissipation associated with the contraction phase of the flow is neglected. This mechanism becomes significant when flow velocity increases, which occurs at lower porosity. The analytical model considers only dissipation due to viscous eddies forming downstream of the screen under elevated velocities. That mechanism is less relevant at lower flow speeds associated with higher porosity. As a result, the analytical formulation tends to overpredict forces at low porosity and underpredict them at high porosity. Furthermore, the analytical model assumes linear waves that does not account for wave asymmetry. In fact, when waves are better approximated by higher-order Stokes (as in the numerical model and in real cases) rather than by linear theory, the crest tends to be higher and sharper, and the trough flatter (Zhang et al., 2022). For this reason, real waves tend to have higher crest velocity and therefore dissipate more through the wave screen gaps. Despite these limitations, the analytical solution remains a useful tool for estimating the forces acting on a wave screen.

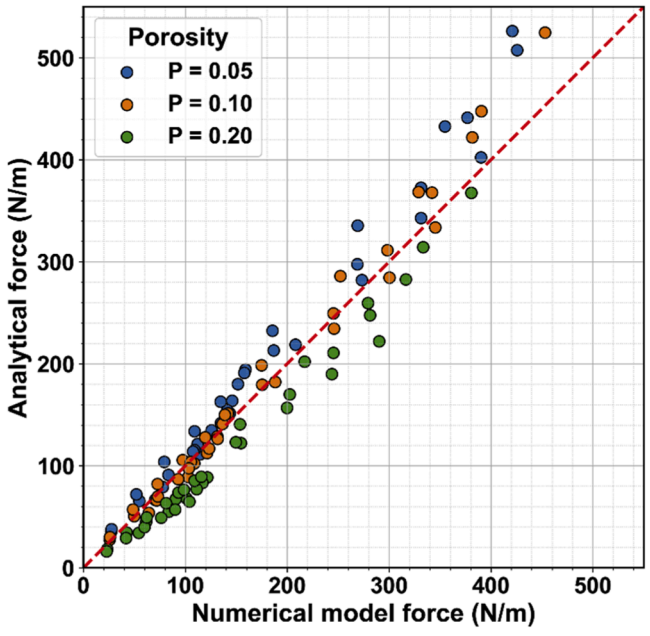


Fig. 13. Numerical model results against linear theory for all the different geometric combinations (as discussed in Section 2.1) (R^2 = 0.92). Note that forces are expressed per linear meter along the width of the structure.

4. Discussions

4.1. Comparison with other studies

Our findings on porosity are consistent with established trends for wave screens. Both experimental and numerical results show that wave transmission decreases and reflection increases as porosity is reduced, in agreement with early theoretical formulations and later laboratory studies (Kriebel, 1993; Alkhalidi et al., 2020). In contrast, the present results indicate that variations in porosity have a limited effect on wave-induced forces. From a design perspective, this suggests that moderate increases in force can be accommodated structurally, while reductions in transmitted wave energy provide a more significant benefit to the protected area.

The strong sensitivity of wave transmission to relative draft observed in this study is also consistent with linear predictions for finite-draft barriers and with closed-form expressions for transmission and reflection coefficients. In particular, for higher wave steepness, minimum transmission occurs at intermediate values of a/h, reflecting a shift predicted by idealized models once viscous and jet-related losses are considered (Suh et al., 2005). In contrast, wave reflection exhibits a much weaker sensitivity to draft, reaching approximately 90% of its maximum value when the wave screen is submerged by only 20% of the total water depth. This behavior can be attributed to the depth-dependent distribution of wave energy in the water column. Under linear wave theory, wave-induced orbital velocities and dynamic pressure decay exponentially with depth, with the majority of wave energy concentrated near the free surface. Consequently, even a shallow screen intercepts the most energetically significant portion of the incident wave field, producing near-maximum reflection with only modest submergence. Further increases in draft extend the screen into regions of progressively lower energy density, yielding diminishing returns in terms of reflected energy. This is consistent with the known dependence of wave-structure interaction on the vertical extent of the obstacle relative to the characteristic decay length of the wave motion. To our knowledge, this rapid saturation of reflection at shallow submergence has not been previously reported.

The monotonic increase in forces with increasing draft observed under steeper wave conditions is compatible with drag-augmented formulations developed for vertical slotted barriers, which similarly predict higher loads as effective blockage and wetted area increase (Poguluri and Cho, 2020). The present results further indicate that these force variations are driven primarily by changes in the incident wave field rather than by geometric parameters alone, emphasizing the importance of wave characteristics in design.

However, the comparisons presented here are limited to regular wave conditions and single-screen configurations. Consequently, the observed relationships between porosity, draft, and hydrodynamic response may not fully capture the behavior under irregular wave fields, multi-directional wave spectra, or multiple-row screen arrangements. These factors should be considered when extrapolating the present results to full-scale coastal or harbor applications.

4.2. Analytical theory

The analytical formulation developed in this study is grounded in linear wave theory and is intended as a simplified interpretive framework rather than a full potential-flow solution of the wave–screen interaction problem. In contrast to more rigorous approaches, such as those of Fugazza and Natale (1992), in which velocity potentials for the incident and reflected waves are obtained by enforcing far-field boundary conditions and matching conditions at the wall boundary, the present model follows the schematic method proposed by Kriebel (1993). This approach is explicitly based on fundamental hydraulic principles and a first-order idealization of the flow, and is therefore designed to support interpretation of numerical and experimental trends

rather than to provide an exact analytical description. Within this scope, the model captures the dominant mechanisms governing wave transmission, reflection, and force generation under regular wave conditions and single-screen configurations.

A primary limitation of the analytical theory arises from its first-order idealization of the wave-induced motion in the vicinity of the screen as a quasi-uniform flow over the analyzed region, effectively corresponding to a long-wave approximation. Within this framework, Eq. (10) assumes energy conservation during the contraction phase of the flow through the screen openings. Energy losses are accounted for only during flow expansion, represented through the formation of viscous eddies downstream of the structure. In reality, dissipation also occurs during flow contraction as orbital velocities accelerate within the gaps of the screen, a process that depends explicitly on the local velocity field and is not represented in the present formulation. Future analytical developments could address this limitation by introducing a dissipation term linked directly to the orbital velocity during contraction.

The analytical model further assumes symmetric wave shapes as a direct consequence of the underlying linear and hydraulic approximation. The neglect of wave asymmetry is therefore not a higher-order correction but an inherent aspect of the first-order formulation. While asymmetric crest–trough behavior in real ocean waves may influence transmission, reflection, and force estimates under steeper or more nonlinear conditions, capturing such effects would require extending the present approach beyond its intended scope to include higher-order pressure and velocity terms or a full potential-flow treatment.

4.3. Practical design of the wave screen

Increasing the submergence of the wave screen and decreasing its porosity both lead to an effective reduction in wave transmission, thereby enhancing protection in the sheltered area. This is particularly relevant for harbor and marina environments, where attenuating wind waves and vessel wakes is essential to prevent structural damage and operational disruptions.

The reduction in wave transmission has a limited influence on wave reflection (Fig. 10), allowing improved protection on the sheltered side while maintaining acceptable reflection levels on the offshore side. This balance is important in harbor settings that require inshore shielding without compromising navigation safety or docking operations. However, while deeper submergence improves attenuation, it also leads to increased wave-induced forces, which must be explicitly accounted for during structural design.

The present analysis does not account for the effects of wave screens on water circulation and flushing within enclosed or semi-enclosed basins. Recent observations in a managed coastal lake showed that reduced flushing can coincide with salinity increase and nitrogen accumulation, emphasizing that wave-screen design should also consider basin renewal and water quality (Biondi et al., 2026). Increased submergence and reduced porosity can restrict flow exchange, potentially leading to stagnation, reduced oxygen levels, and the accumulation of sediments or pollutants, with negative implications for water quality and ecological health (Lee et al., 2018). Quantifying these processes requires coupled hydrodynamic and transport simulations, which are beyond the scope of the present study. In addition, the results are limited to regular wave conditions and single-screen configurations. The response of wave screens to irregular or multi-directional wave fields, as well as the interaction between multiple rows of screens, may alter both attenuation performance and force distribution. The influence of screen spacing on transmission, reflection, and structural loading also remains unresolved. Addressing these aspects would require dedicated numerical and experimental investigations and represents an important direction for future research.

5. Conclusions

This study investigated the interaction of regular waves with a partially submerged wave screen using laboratory experiments, OpenFOAM based RANS simulations, and analytical modeling. Beyond reproducing wave transmission and reflection, the work explicitly quantified wave induced forces and linked them to changes in porosity, submergence, and wave characteristics.

A key contribution of this study is the demonstration that porosity strongly controls wave transmission and energy dissipation while exerting only a secondary influence on forces. For the tested configurations, reducing porosity by 15% increased forces by approximately 10% while enhancing energy dissipation by more than 20%, indicating that substantial gains in protection can be achieved with limited structural penalty. In addition, it was shown that approximately 90% of the maximum reflection is achieved at a relative draft of only 0.2, highlighting diminishing returns of deeper submergence in terms of wave reflection.

Another novel finding is the dominant role of wave steepness in governing both attenuation and loading. High steepness waves lead to increased dissipation and forces even under shallow submergence, and produce plateaus in transmission, reflection, and force growth at higher relative drafts. This behavior is not captured by simplified design assumptions based solely on wave height or period.

An analytical formulation was developed to interpret these results and isolate the role of nonlinear and viscous effects, which become significant for low porosity screens and steeper waves. The discrepancies between analytical predictions and numerical results provide insight into the limits of linear theory for slotted barriers.

The combined experimental, numerical, and analytical framework presented here advances current understanding by jointly addressing wave attenuation and structural loading, and by identifying wave steepness as a governing design parameter. These results provide practical guidance for optimizing wave screen designs that balance sheltering efficiency, force demand, and water exchange in coastal and harbor environments.

CRediT authorship contribution statement

Stefano Biondi: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis. **Alberto Canestrelli:** Writing – review & editing, Supervision, Resources, Methodology, Conceptualization. **Chiara Favaretto:** Writing – review & editing, Methodology, Investigation. **Luca Martinelli:** Writing – review & editing, Methodology. **Piero Ruol:** Writing – review & editing, Methodology. **Edwin Rajeev:** Software, Methodology. **Jason Cummins:** Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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